FACULTEIT GENEESKUNDE EN GEZONDHEIDSWETENSCHAPPEN

SKIN OF COLOR

HOORCOLLEGE DERMATOLOGIE 3DE BACHELOR GENEESKUNDE

UNIVERSITEIT GENT

Dr. Isabelle Hoorens
Staflid Dermatologie

INHOUD LES

1. Definitie, epidemiologie en culturele aspecten
2. Biologie, structuur en functie van de skin of color
3. Meest frequente pathologie in SOC
4. Fysiologische veranderingen

DEEL 1: DEFINITIE, EPIDEMIOLOGIE EN CULTURELE ASPECTEN

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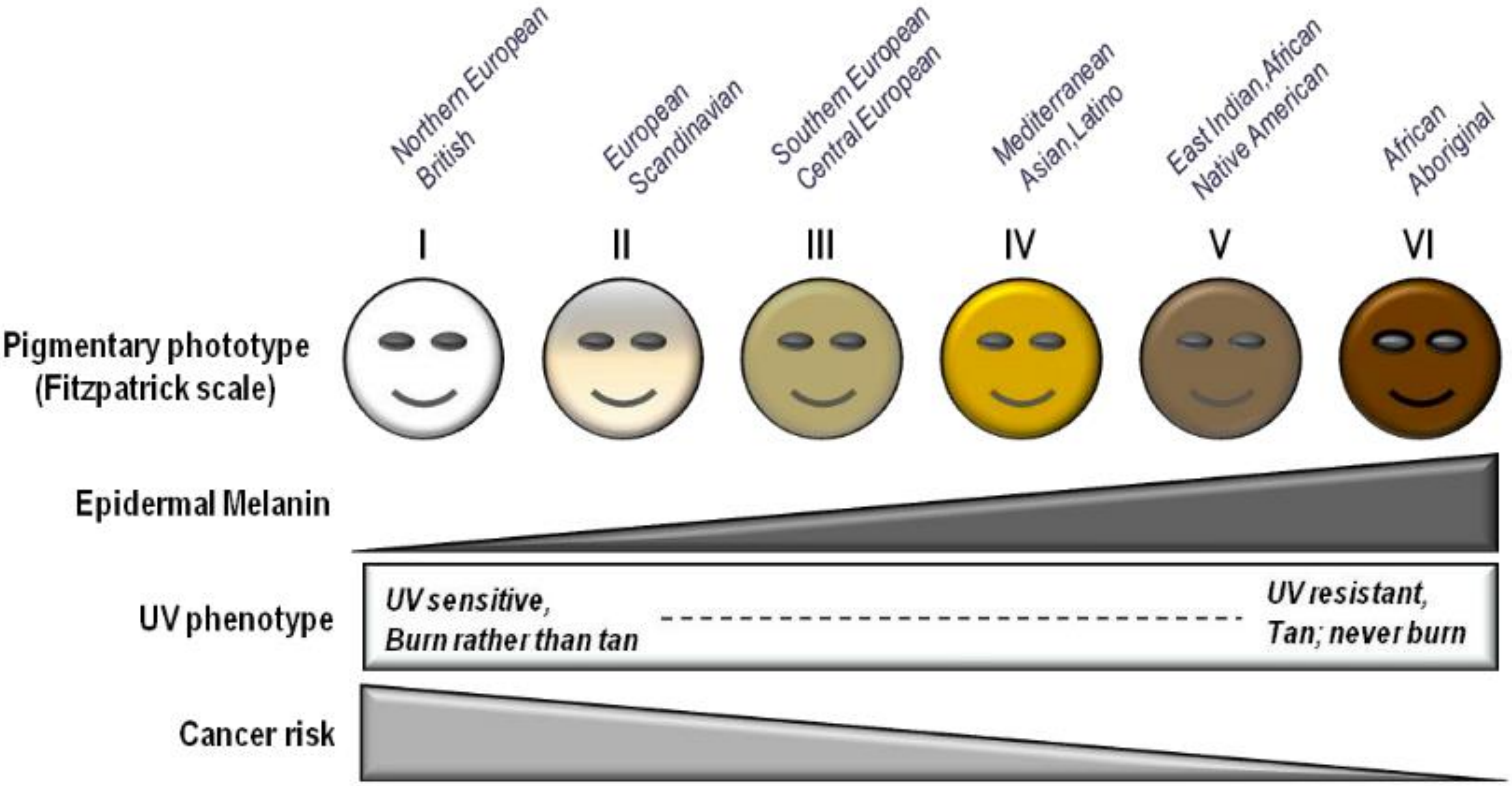
Originator of theory	Date of theory	Theory on skin of color
Native American Indians	Exact date unknown	The creator had not yet perfected his cooking technique while creating humans, burning or undercooking the first humans and thereby causing differences in skin color. Early men jumped into a body of water, progressively dirtying the water and coloring men's skin darker and darker.
African tribes	Exact date unknown	Distribution of the meat of an ox resulted in differing skin colors in those who ate different parts.
Ancient Greeks	c. 1500–400 BC	Phaeton flew the sun chariot too close to or too far from the earth, burning black those to whom he flew too close, and turning others pale when he flew too far away.
Abrahamic religions	c. 1400 BC	Cain slew Abel in jealousy over God's favor, and God put a dark mark on Cain and all of his descendants as punishment. Ham was cursed with blackness because he disobeyed the prohibition against sexual intercourse aboard the ark. Noah cursed Ham when he looked upon his father's nakedness, resulting in a black mark of shame on all of Ham's descendants. Gehazi, servant of Elisha, was cursed with leprosy and the resulting white skin for having solicited money from Naaman.
Leonardo da Vinci	c. 1470 AD	Humans' different skin colors could be explained by the differences in environment.
Paracelsus	1520	Black and white people (termed the children of Adam) had entirely separate origins.
Isaac de La Peyrère	1655	Man descended from either an Adamite or a pre-Adamite race, with natives of Africa, Asia, and the New World being descended from pre-Adamites.
Johann F. Blumenbach	1795	Differences in human skin color were produced by a combination of climate and other factors. Black colored skin was thought to result from embedded carbon due to the heat of tropical climates.
Samuel Stanhope Smith	1810	Skin color was attributable mainly to climate, but skin color could eventually become hereditary.
Benjamin Rush	1812	Black skin was a hereditary illness.
Johann Meckel	1816	Skin color emanated from the cortical part of the brain.
Julien-Joseph Virey	1837	Based on observations of changes in newborns' skin color, it was determined that external factors had little or no effect on skin color.
Joseph Smith Jr.	1840	The Lamanites had once been white but were cursed by God and turned black.

Robert Chambers	1844	Nonwhite people were in the process of developing to the highest race as Caucasians.
Samuel G. Morton	1847	Black and white people were not varieties of a single race but entirely different species; this was based on the conviction that "half-breeds" cannot propagate themselves indefinitely.
Charles Darwin	1871	The theory of evolution left no doubt that all humans belong to the same species.
Rudolph Matas	1896	The chemical composition of pigment in black skin was identical to white skin.
MODERN SCIENTIFIC THEORIES OF SKIN OF COLOR		
Vitamin D/sunlight theory		
Charles Loring Brace IV	1987	The need for protection from ultraviolet radiation leads to darker skin color.
Thomas B. Fitzpatrick	1988	The Fitzpatrick skin type scheme classifies skin types I to VI by the response of the skin to sun exposure, in terms of the degree of burning and tanning of the skin. However, many dermatologists have adopted Fitzpatrick's skin type classifications without correlating the amount of sun exposure to the skin type category.
Nina Jablonski and George Chaplin	2010	Skin pigmentation is the result of natural evolutionary processes trying to balance between protecting the skin from harmful ultraviolet rays and the absorption of vitamin D.
Biology of the melanin pigmentary system		
George Szabó	1959	The use of light and electron microscopes led to the understanding that melanocytes are symmetrically distributed and do not differ significantly in size, shape, or population density between the various races.
Walter Quevedo Jr.	1971	Pigmentation of the skin is related to the formation and melanization of melanosomes in melanocytes. The amount of melanin present in keratinocytes is the essential factor in determining skin color.
Genetics and the DNA of skin of color		
Human Genome Project (HGP)	1990–2003 and ongoing	The HGP is helping redefine who humans are and how we have evolved. This project may help us answer questions about the evolution of the color of skin. The mapping of DNA has proven that human beings are genetically the same and that skin color is simply a variation.

Source: Used with permission from Dr. Beverly Baker-Kelly, Dr. Mouhiba Jamoussi, Ms. Natasha Savoy Smith, and Ms. Rachel Schiera.

INDELING HUIDTYPES

- Fitzpatrick (1988): semi-kwantitatieve schaal
- Huidskleur gekwantificeerd op basis van
  - Basaal complex
  - Melanine niveau
  - Inflammatoir antwoord op UV of MED: minimal erythematous dose = hoeveelheid UV (voornamelijk UVB) nodig om zonnebrand te induceren na 24–48 h)
- Risico op huidkanker
- 6 Fitzpatrick huidtypes

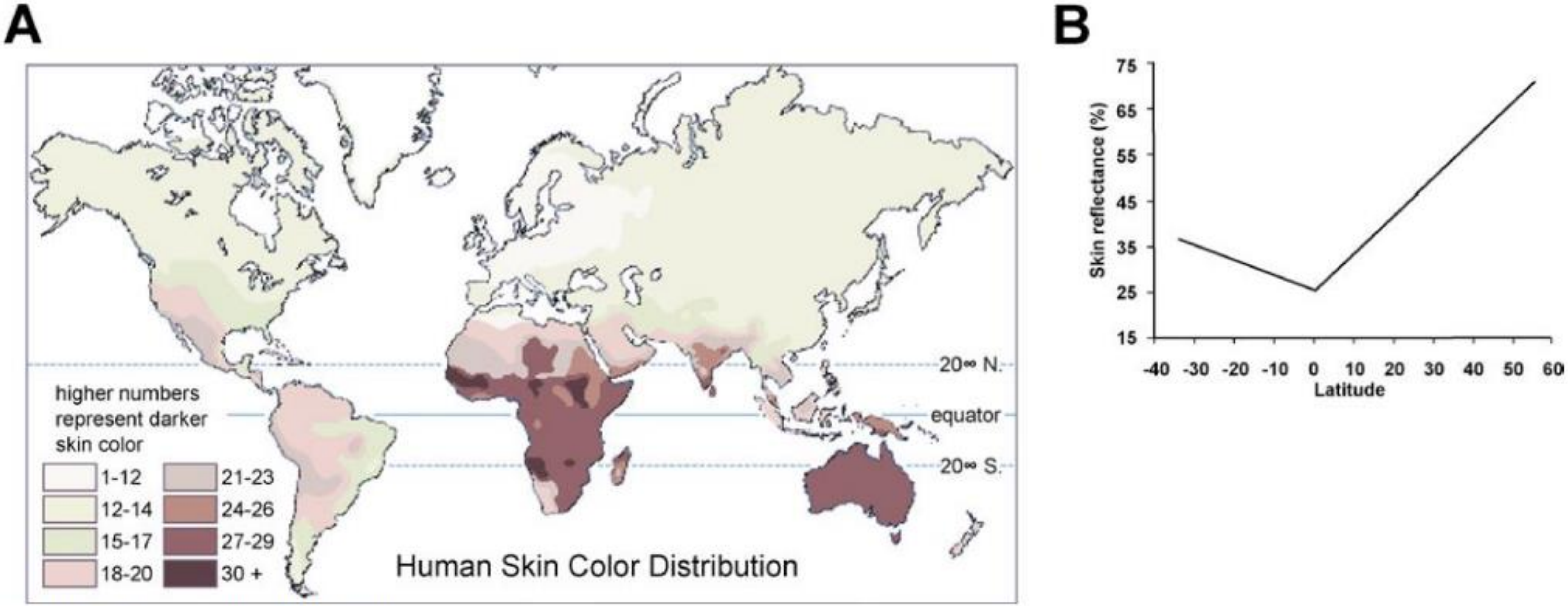


Northern European
British
European
Scandinavian
Southern European
Central European
Mediterranean
Asian, Latino
East Indian, African
Native American
African
Aboriginal
I
II
III
IV
V
VI
Pigmentary phototype
(Fitzpatrick scale)
Epidermal Melanin
UV phenotype
UV sensitive,
Burn rather than tan
UV resistant,
Tan; never burn
Cancer risk

Table 1. Skin pigmentation, the Fitzpatrick scale and UV risk.

Fitzpatrick phototype	Phenotype	Epidermal eumelanin	Cutaneous response to UV	MED (mJ/cm^2) *	Cancer risk
I	Unexposed skin is bright white Blue/green eyes typical Freckling frequent Northern European/British	+/−	Always burns Peels Never tans	15–30	++++
II	Unexposed skin is white Blue, hazel or brown eyes Red, blonde or brown hair European/Scandinavian	+	Burns easily Peels Tans minimally	25–40	+++/++++
III	Unexposed skin is fair Brown eyes Dark hair Southern or Central European	++	Burns moderately Average tanning ability	30–50	+++
IV	Unexposed skin is light brown Dark eyes Dark hair Mediterranean, Asian or Latino	+++	Burns minimally Tans easily	40–60	++
V	Unexposed skin is brown Dark eyes Dark hair East Indian, Native American, Latino or African	++++	Rarely burns Tans easily and substantially	60–90	+
VI	Unexposed skin is black Dark eyes Dark hair African or Aboriginal ancestry	+++++	Almost never burns Tans readily and profusely	90–150	+/−

Minimal erythematous dose (MED) is defined as the least amount of UVB radiation that causes reddening and inflammation of the skin 24–48 h after exposure (*i.e.*, the lowest UV dose that causes sunburn). The more UV sensitive an individual is, the lower the MED of his/her skin.



DOI: 10.1371/journal.pbio.0000027.g002

Figure 2. Relationship of Skin Color to Latitude
(A) A traditional skin color map by Biasutti, based on http://anthro.palomar.edu/vary/. (B) Summary of 102 skin reflectance samples for males as a function of latitude, redrawn from Relethford (1997).

DEFINITIE

- Skin of color = individuen van raciale groepen met meer donkere huid dan Kaukasiërs zoals Aziaten, Afrikanen en Indiaanse populatie
- Patiënten met een donkere huid hebben vaak specifieke karakteristieken zowel betreffende hun huid, haren, dermatosen, reactiepatronen alsook specifieke culturele aspecten met weerslag op hun huidzorg
- Er zijn huidziekten, genetische aspecten, en reacties of huidafwijkingen op trauma als gevolg van het huidtype
- De top dermatologische diagnosen in patiënten met skin of color zijn alopecia, keloïden en seborrhoische dermatitis

Stevens - Johnson syndrome

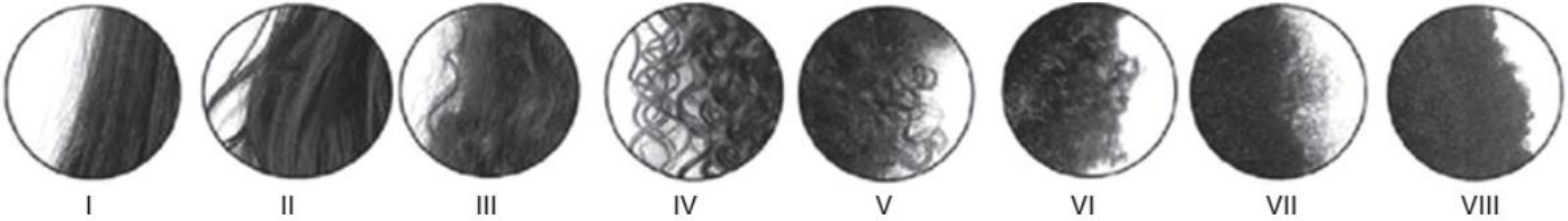
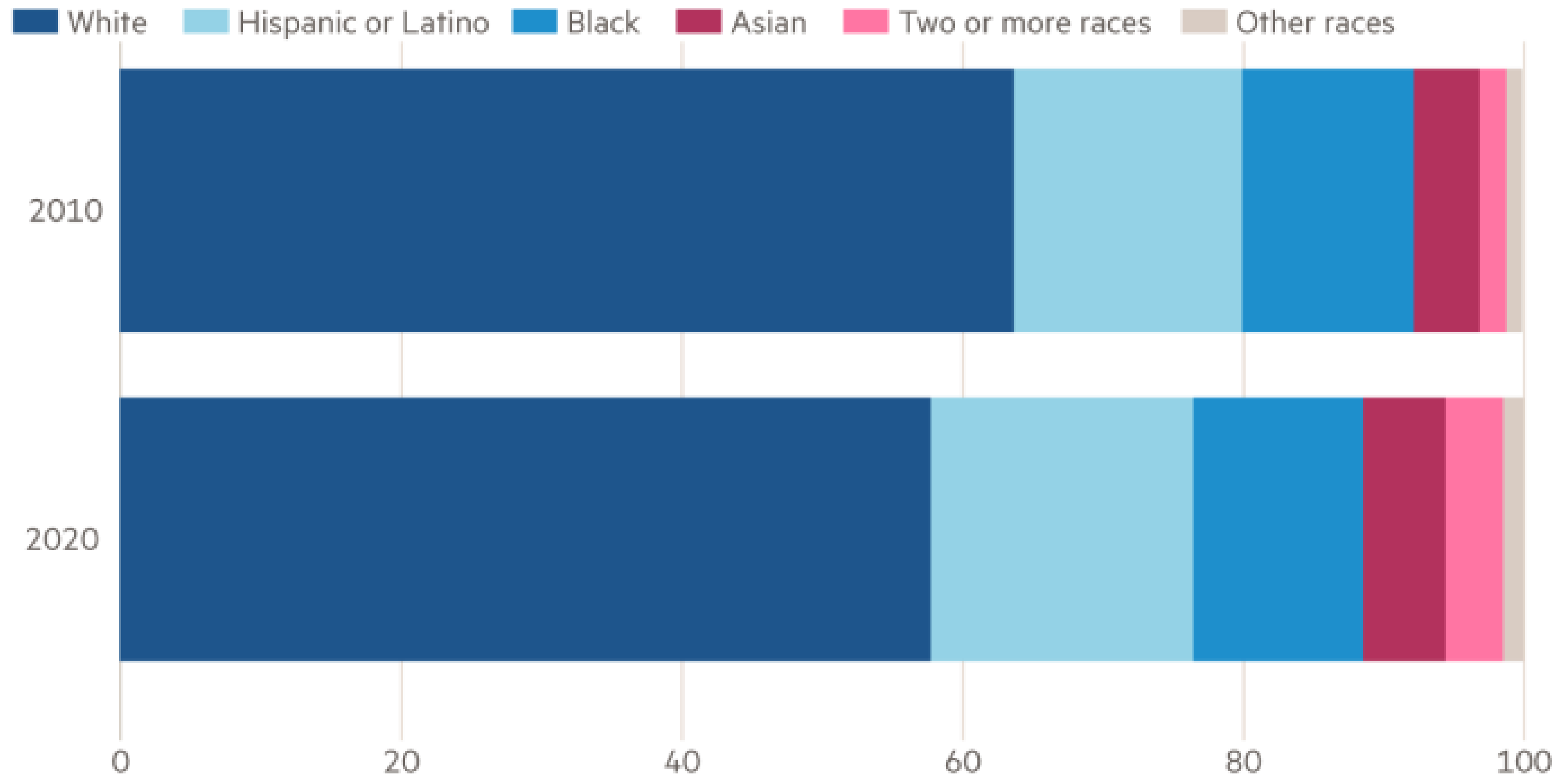


FIGURE 2-2. The eight types of human hair in terms of degree and type of curl, as described by Loussouarn et al. (Reproduced with permisison from Loussouarn G, Garcel AL, Lozano I, et al: Worldwide diversity of hair curliness: a new method of assessment, *Int J Dermatol.* 2007 Oct;46 Suppl 1:2-6).[5a]

BELANG

More than 4 in 10 Americans identified as non-white in the 2020 census

Share of population (%)



Race categories do not include those who are Hispanic or Latino
Source: US Census Bureau
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GETUIGENISSEN SAMENLEVING

‘Je kunt zwarte mensen niet meer wegdenken uit de Belgische samenleving’

De Standaard vroeg vijf Belgen met Afrikaanse roots naar hun kijk op het belang van geschiedenis en gemeenschap. ‘Mensen zien eerst die kleur, en dan pas de rest.’

Josephine Dapaah

De Standaard, 1 maart 2022

CULTURELE ASPECTEN

- Kruiden (chinese geneeskunde) zoals arnica extracten veroorzaken sweet syndroom
- Haarzorg (afro-amerikaanse populatie) veroorzaken tractie-alopecia
- Spaanse populatie heeft verschillende huis en tuin remedies voor de meest gangbare dermatologische aandoeningen (visolie voor psoriasis en melk voor zonnebrand)
- Prayers nodules in moslimculturen door frequente repititieve houding tijdens het bidden

DEEL2: BIOLOGIE, STRUCTUUR EN FUNCTIE VAN DE SKIN OF COLOR

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WAT VEROORZAAKT DE HUIDSKLEUR?

- Huidskleur: pigmentatie (melanine)
- Aantal en functie melanocyten gelijk over de huidtypes
- Melanine onstaat uit melanocyten
- Melanocyten geven pigmentatie af via melanosomen
  - Melanosomen: voorverpakte organellen
- Melanine
  - Natuurlijke bescherming tegen UV
  - Samengesteld uit verschillende pigementen
  - < tyrosine (aminozuur)
  - Verschillende andere rollen naast UV protectie

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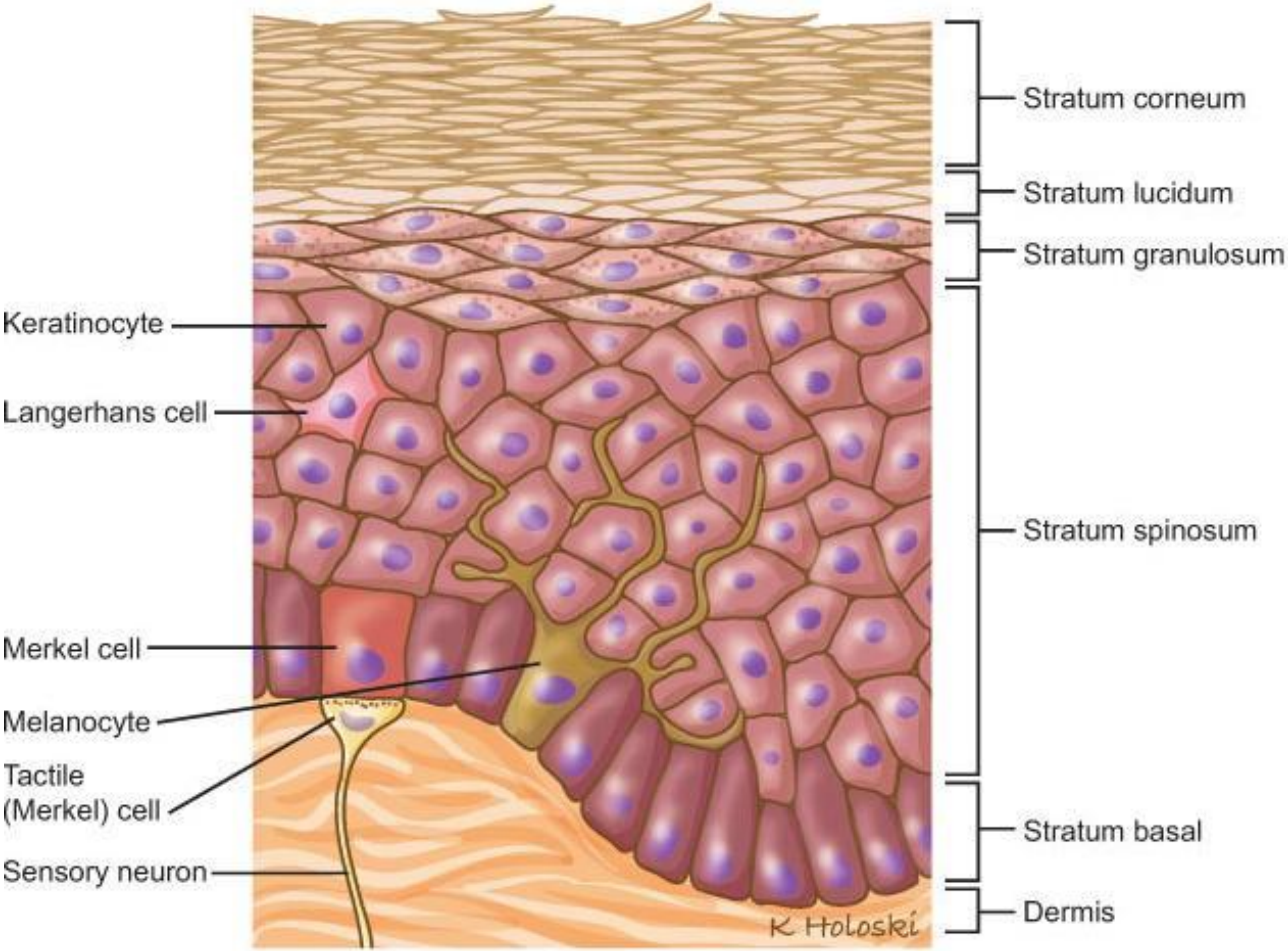
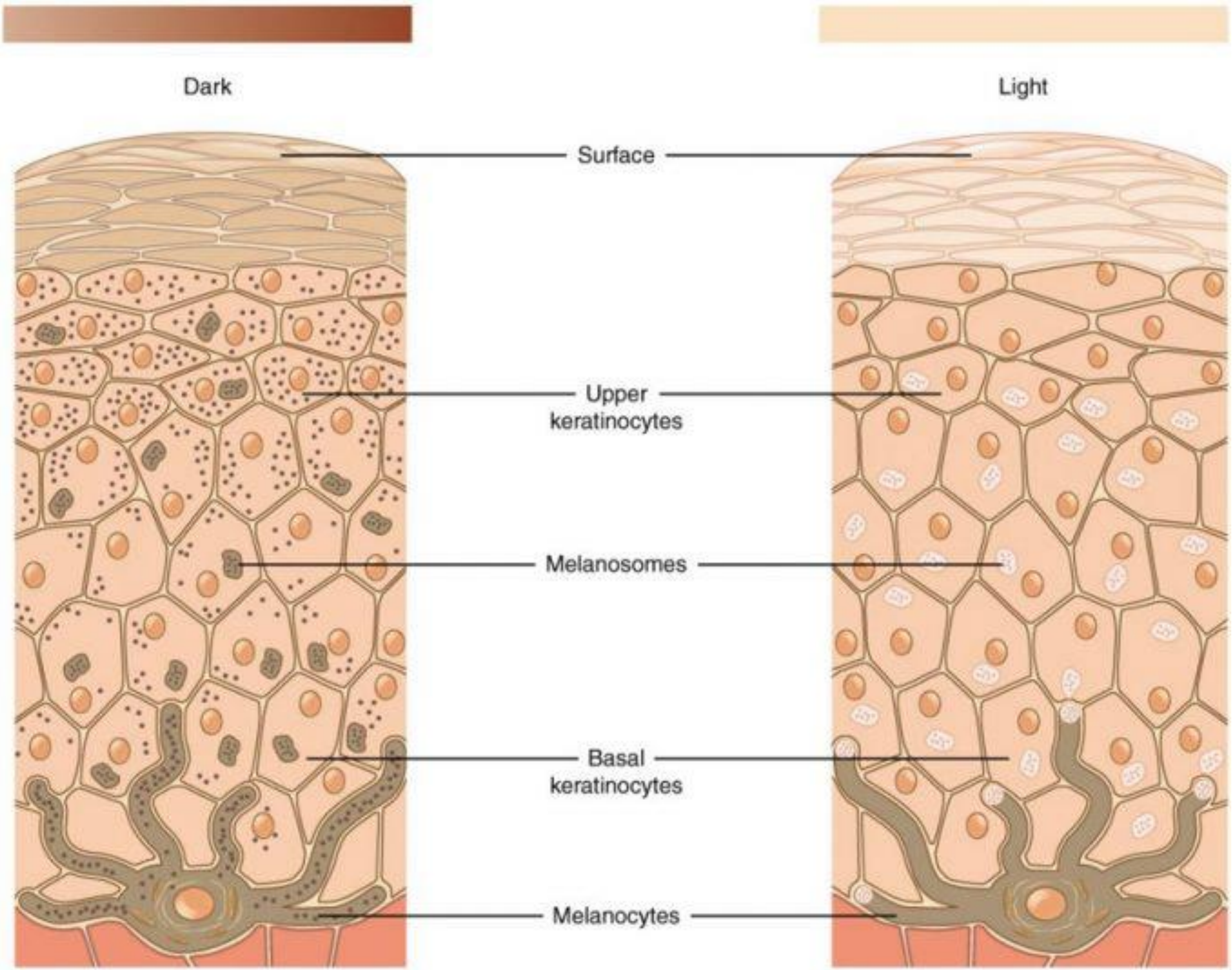

Stratum corneum
Stratum lucidum
Stratum granulosum
Keratinocyte
Langerhans cell
Stratum spinosum
Merkel cell
Melanocyte
Tactile
(Merkel) cell
Sensory neuron
Stratum basal
Dermis
K Holoski



Dark
Light
Surface
Upper keratinocytes
Melanosomes
Basal keratinocytes
Melanocytes

- Huidskleur = melanine bestaande uit 2 chemische samenstellingen
- Eumelanine
  - Donker: bruin/zwarte melanine
  - Blokkeert UV fotonen zeer goed
- Pheomelanine
  - Lichter: rood/blond melanine
  - Minder effectieve UV bescherming
  - Vrije radicalen die aanleiding geven tot DNA schade (melanoma)

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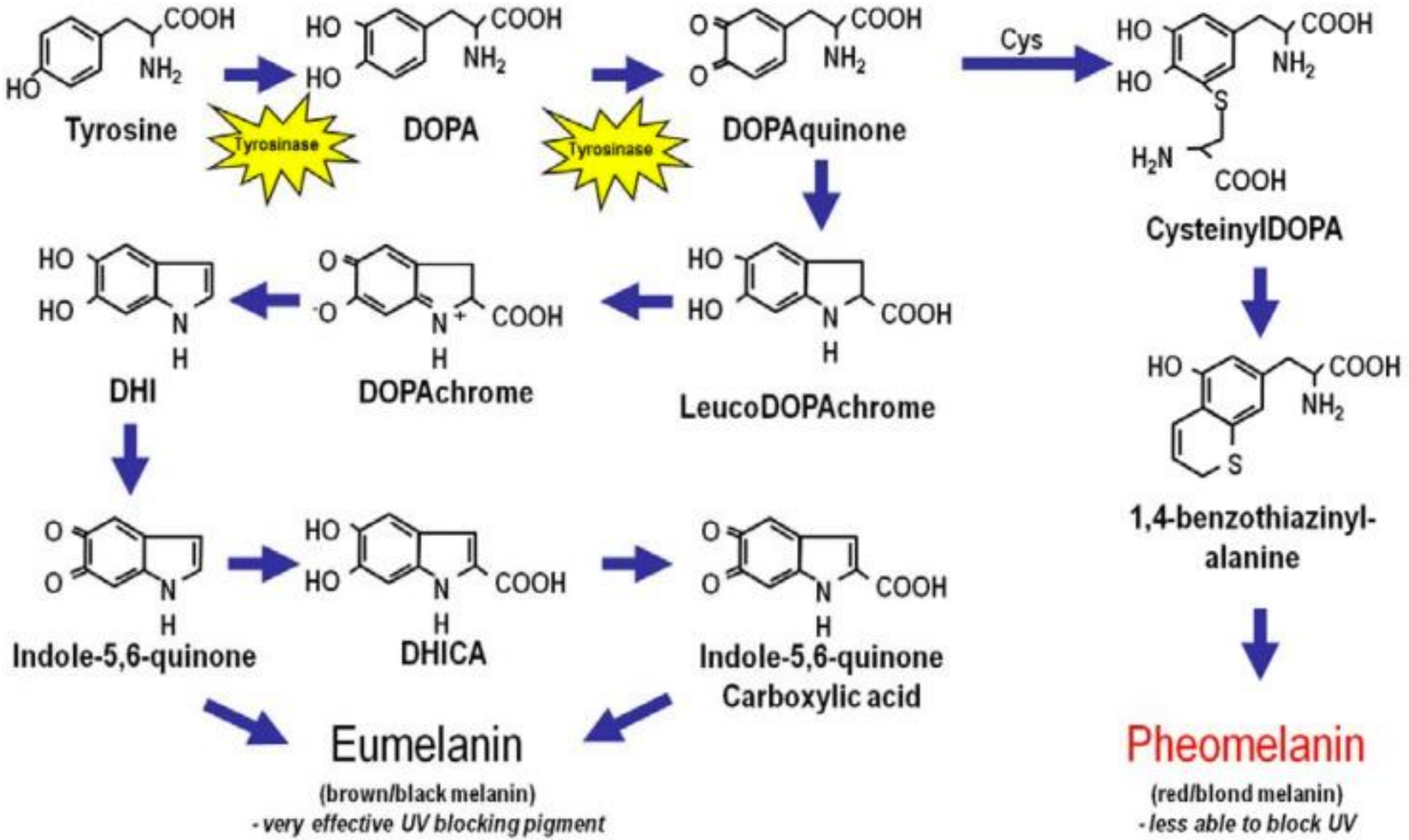
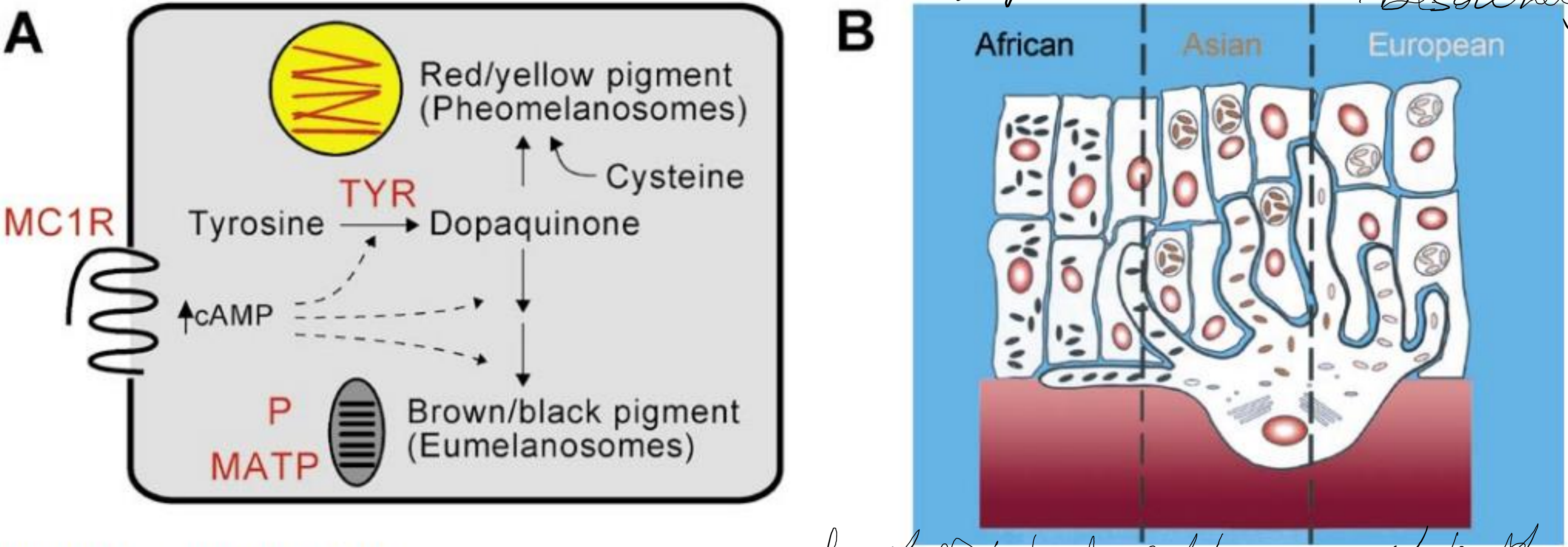


Tyrosine
Tyrosinase
DOPA
Tyrosinase
DOPAquinone
Cys
CysteinylDOPA
DHI
DOPAchrome
LeucoDOPAchrome
1,4-benzothiazinyl-alanine
Indole-5,6-quinone
DHICA
Indole-5,6-quinone Carboxylic acid
Eumelanin
(brown/black melanin)
- very effective UV blocking pigment
Pheomelanin
(red/blond melanin)
- less able to block UV

vnl UVB geeft aanleiding tot mutaties
in basale laag



fysiologisch bruin verkleuren v/d huid brengt per definitie al DNA-schade met zich mee

DOI: 10.1371/journal.pbio.0000027.g001

Figure 1. Biochemistry and Histology of Different Skin Types
(A) Activation of the melanocortin 1 receptor (MC1R) promotes the synthesis of eumelanin at the expense of pheomelanin, although oxidation of tyrosine by tyrosinase (TYR) is required for synthesis of both pigment types. The membrane-associated transport protein (MATP) and the pink-eyed dilution protein (P) are melanosomal membrane components that contribute to the extent of pigment synthesis within melanosomes. (B) There is a gradient of melanosome size and number in dark, intermediate, and light skin; in addition, melanosomes of dark skin are more widely dispersed. This diagram is based on one published by Sturm et al. (1998) and summarizes data from Szabo et al. (1969), Toda et al. (1972), and Konrad and Wolff (1973) based on individuals whose recent ancestors were from Africa, Asia, or Europe.

THE HUMAN GENOME PROJECT

- HGP helpt om antwoorden te vinden betreffende de evolutie van de skin of color
- DNA onderzoek toont aan dat alle mensen in genetica gelijk zijn, onafhankelijk van oppervlakkige variaties in huidskleur of textuur
- Mapping van genen en DNA in het HGP kan de dermatologen helpen om variatie in dermatologische aandoeningen te verklaren en verder onderzoek op te zetten

STRUCTUURVERSCHILLEN

- Epidermale verschillen
  - stratum corneum structuur
  - lipiden samenstelling
  - melanine samenstelling.
- Dermale verschillen
  - structurele organisatie van de dermis
  - verschillende concentratie van dermale componenten
- Biologische verschillen in structuur en functie resulteren in
  - lagere aantallen huidkankers
  - minder uitgesproken photo-aging (pigmentatie, elastose en rimpels)
  - aanwezigheid van keloïd (grotere fibroblasten)
  - meer pigment afwijkingen

Characteristic	Caucasian descent	African descent	Asian descent
Stratum corneum thickness	Equal	Equal	
Stratum corneocyte size	Equal	Equal	Equal
Stratum corneum layers	Less	More	Less
Stratum corneum lipids	Low	High	
Ceramide concentration	High	Low	High
Melanin	Low	High	Intermediate
Melanosomes	Small, aggregated	Large, dispersed	Mixed
Melanocyte number ← gezn	Same	Same	Same
Melanosome distribution	Stratum basale	Entire epidermis	Mixed
Vitamin D production	High	Low	Intermediate
Minimal erythema dose	Low	High	Intermediate
Photodamage	High	Minimal	Intermediate
Glutathione (reduced state)	High	Low	
Glutathione reductase	High	Low	

DEEL 3: CUTANE ZIEKTE-IMPLICATIES

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REDEN OM DE DERMATOLOOG TE CONSULTEREN

	Reasons by race/ethnicity				
Rank order	African American	Asian/Pacific Islander	White	Hispanic/Latino	Non-Hispanic
1	Dermatitis	Dermatitis	Acne	Dermatitis	Dermatitis
2	Acne	Acne	Dermatitis	Acne	Acne
3	Dermatophytosis	Atopic dermatitis	Actinic keratosis	Cysts	Actinic keratosis
4	Cysts	Urticaria	Viral warts	Viral warts	Viral warts
5	Cellulitis/abscess	Psoriasis	Cysts	Cellulitis/abscess	Cysts
6	Atopic dermatitis	Viral warts	Nonmelanoma skin cancer	Psoriasis	Nonmelanoma skin cancer
7	Candidiasis	Rash/skin eruption	Benign neoplasm	Rash/skin eruption	Benign neoplasm
8	Rash/skin eruption	Cellulitis/abscess	Psoriasis	Scabies	Psoriasis
9	Dermatophytosis	Benign neoplasm	Unspecified skin disorder	Urticaria	Unspecified skin disorder
10	Keloid scar	Cysts	Cellulitis/abscess	Atopic dermatitis	Cellulitis/abscess

Source: Data from Davis SA, Narahari S, Feldman SR, et al. Top dermatologic conditions in patients of color: an analysis of nationally representative data. *J Drugs Dermatol.* 2012 Apr;11(4):466-473.

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ACNE IN SKIN OF COLOR

icepick littekens

papels

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acne rosacea

nodulaire gezwollenheid
v. erythem

DERMATOSIS PAPULOSA NIGRA (DPN) ~ benigne

– 40% van de Afro-amerikaanse bevolking boven de leeftijd van 30 jaar, tevens in andere groepen SOC

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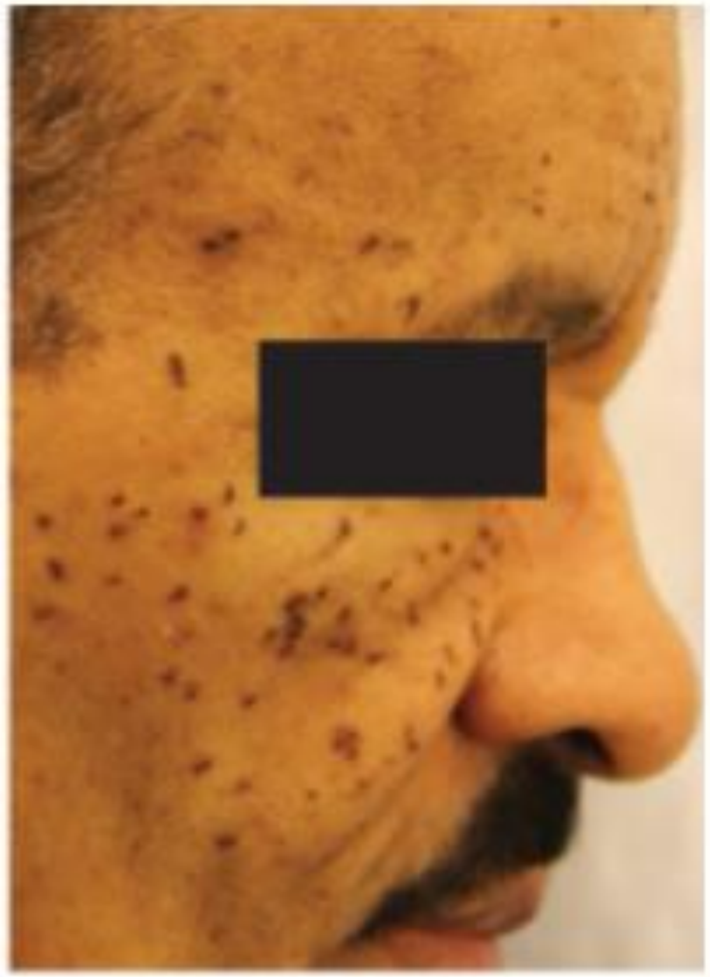
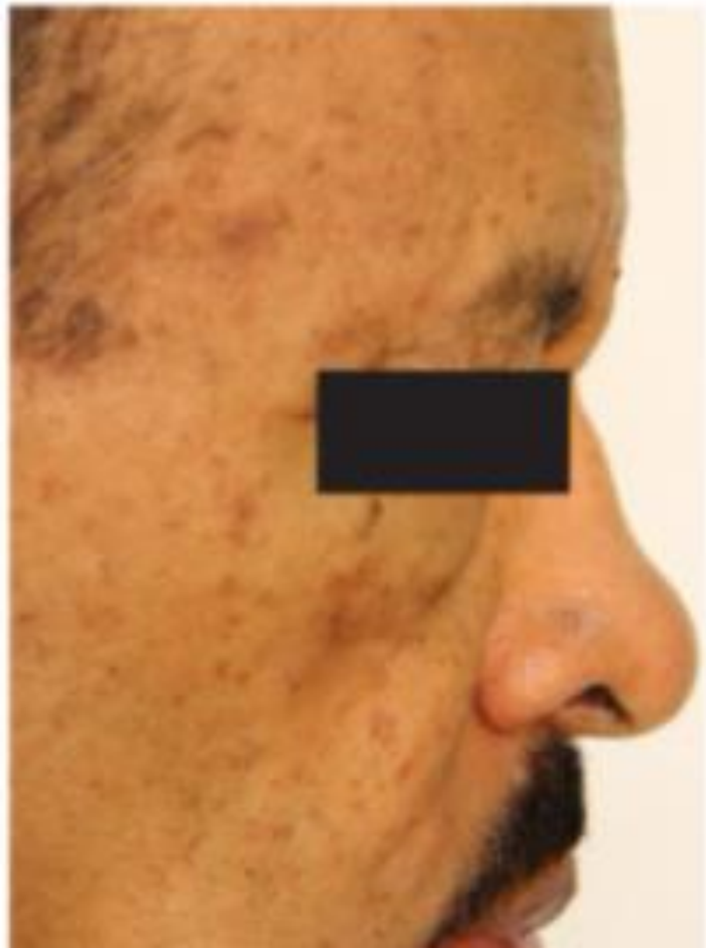
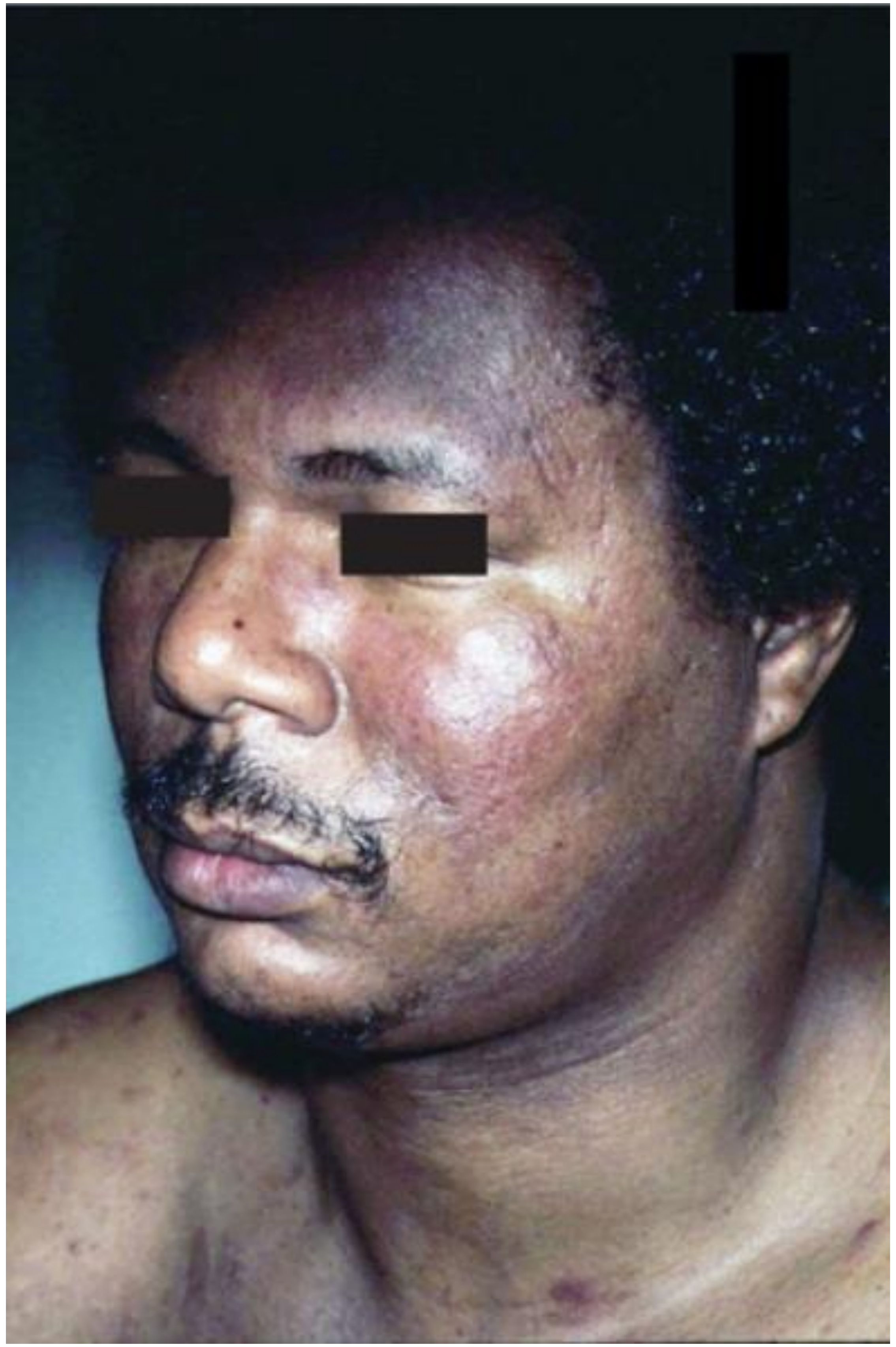
A


B



SEBORROISCHE DERMATITIS



fijnschilferende eczematoze
randbeving
~ redelijk gelijkaardig uitzicht in andere huidtypes

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TABLE 23-2 Clinical patterns of seborrheic dermatitis

Patient group	Pattern/location
Infants	'Cradle cap' and 'napkin' dermatitis
Adults	Areas with facial hair (such as the eyebrows, mustache, and eyelid margins)
General	Skin flexures (such as the nasolabial folds, postauricular folds, axillae, and groin)
HIV/AIDS	Extensive adult clinical findings, blepharitis, and superinfections

Abbreviations: AIDS, acquired immunodeficiency syndrome; HIV, human immunodeficiency virus.

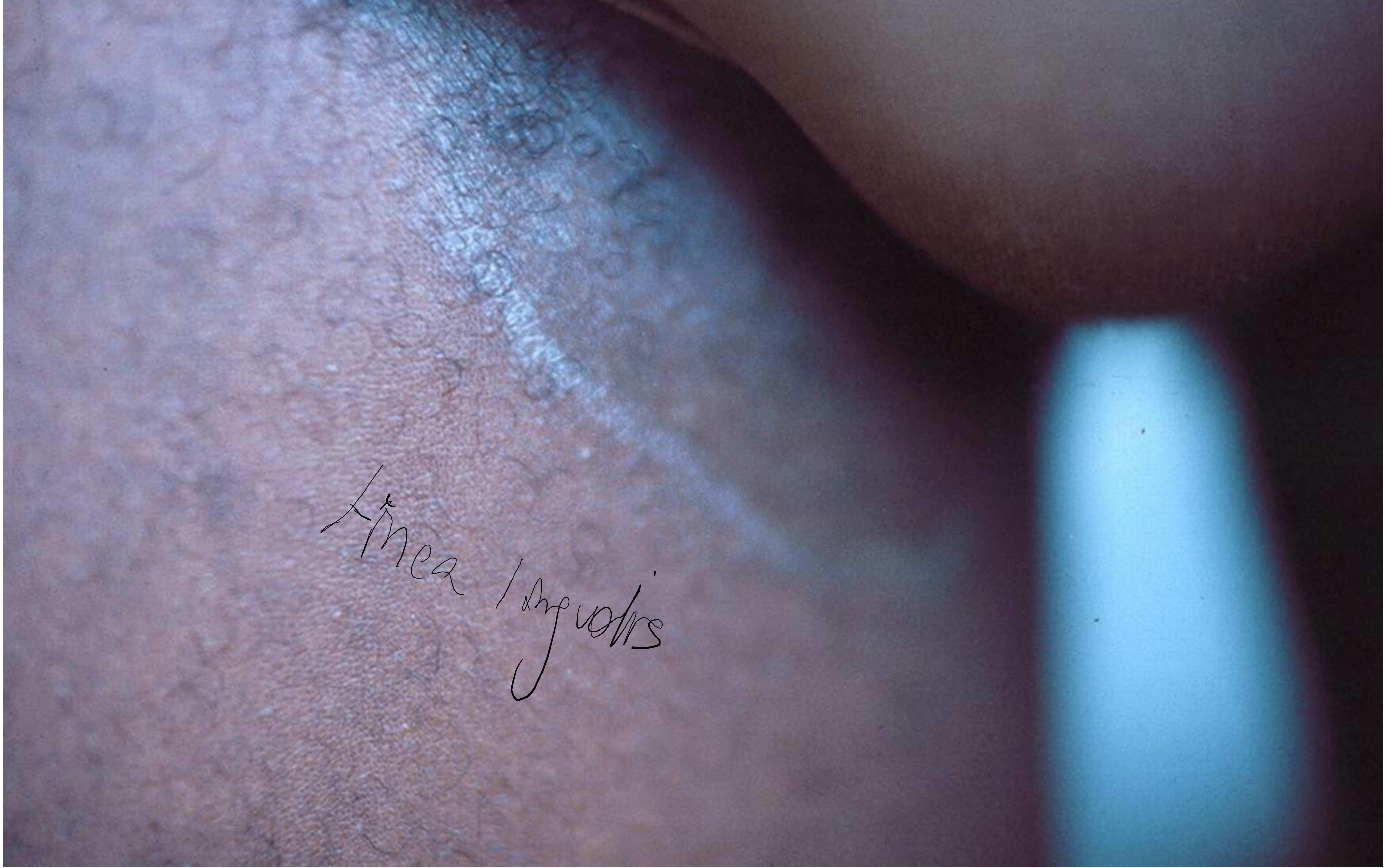
A

B

MYCOSE

verheven actieve rand

schilferende erythemateuze rand


Linea languolis

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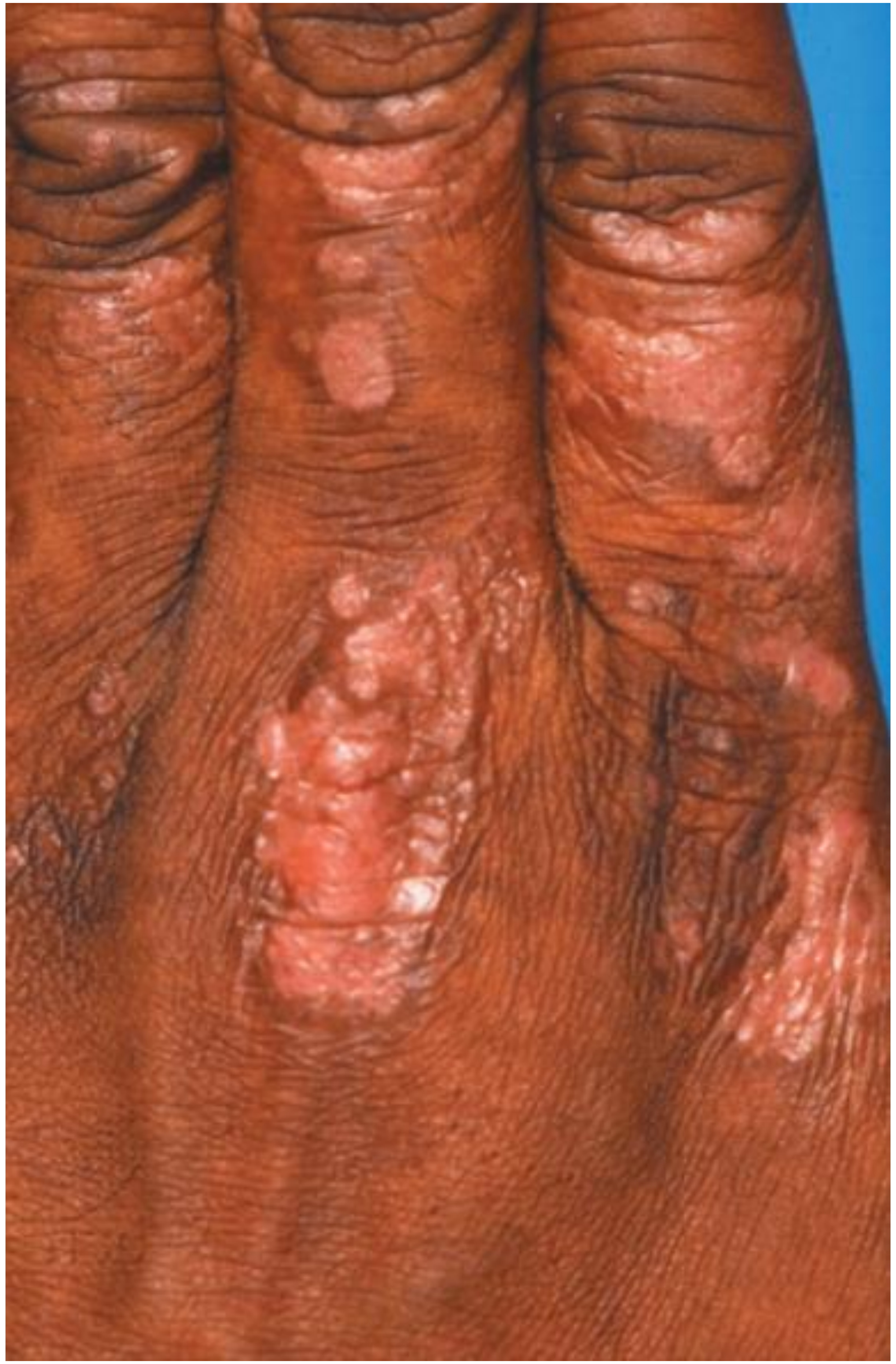
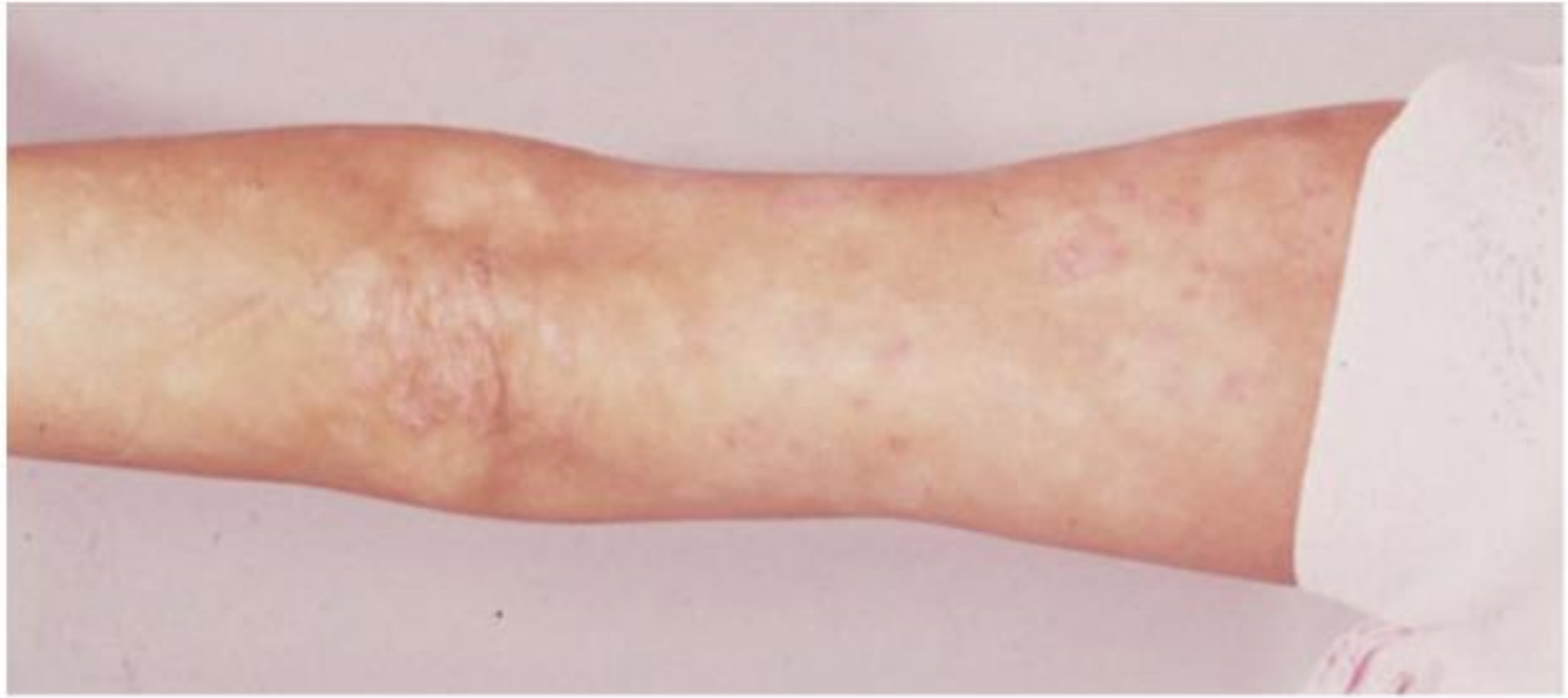
PSORIASIS

- Zeer zeldzaam in de (native) populatie van Zuid Amerika
- Afwezig in (Samoan en Australische) aboriginal populaties
- Nigeria, Angola, Mali en Senegal prevalentie van 0.3% tot 0.8% (lager dan Europese prevalentie)
- In Noord, Zuid en Oost Afrika vergelijkbare prevalenties als Europa 1.3% tot 3%.



psoriatische
handen

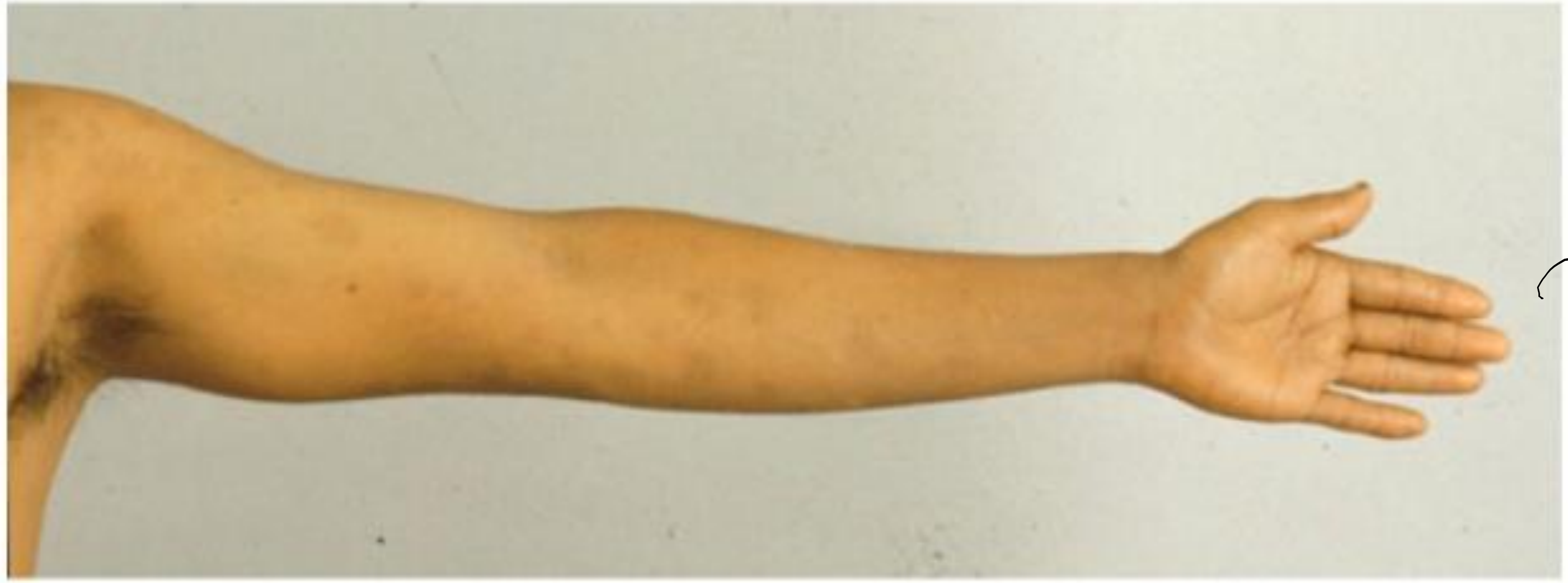
Lichaamsplooien



~ hypopigmentatie

A

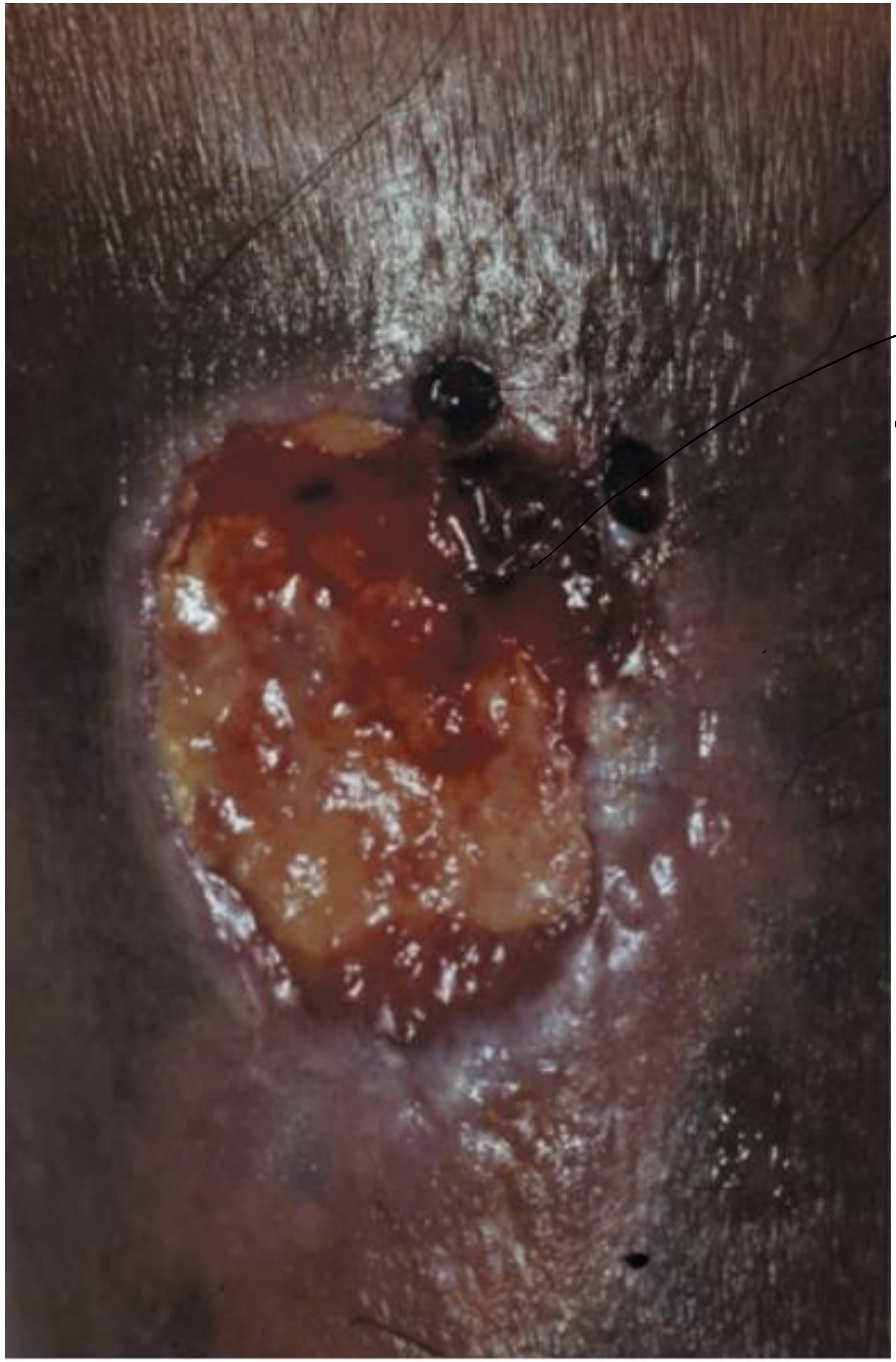
BIJ GENEZING



~ hyperpigmentatie

B

EROSIEVE LETSELS EN WONDEN



Ulcus:
moeilijk te
interpreteren

zwarte ulcer
~ hemosiderine depositie
⇒ teken v. surinfectie

KELOÏDEN

~ door verhoogde aanwezigheid fibroblasten

recidiven
na chirurgie
→ alternatieven Uho

PITYRIASIS ROSEA

A

B

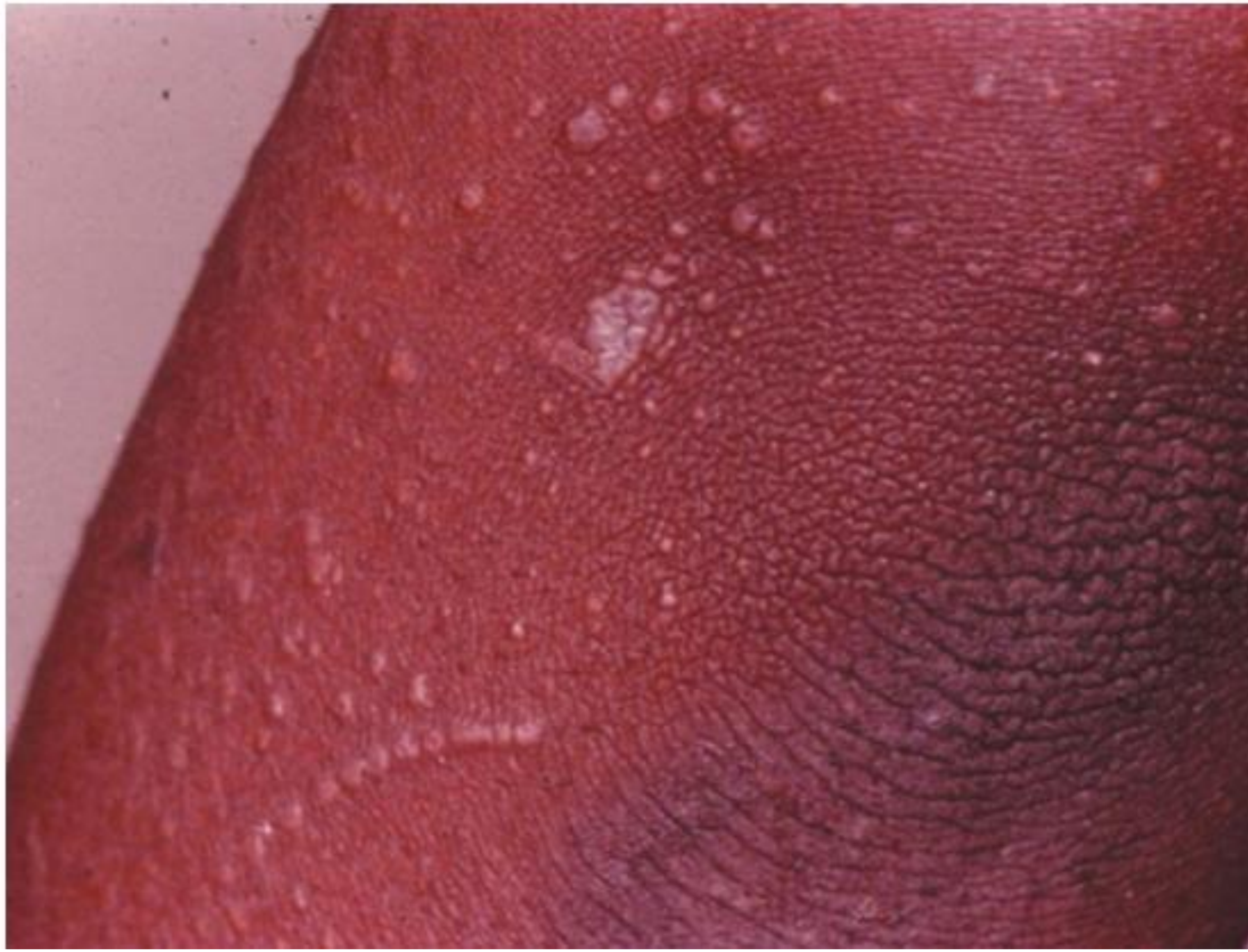
Characteristics	I–III	IV–VI
Color of lesions	Rose or salmon	Gray or violaceous
Location	Predominance truncal	Predominance on the extremities
Scale	Collarette scales	Central scales
Special features	Central hyperpigmentation seldom seen	Central necrotic-like hyperpigmentation

B

A

A

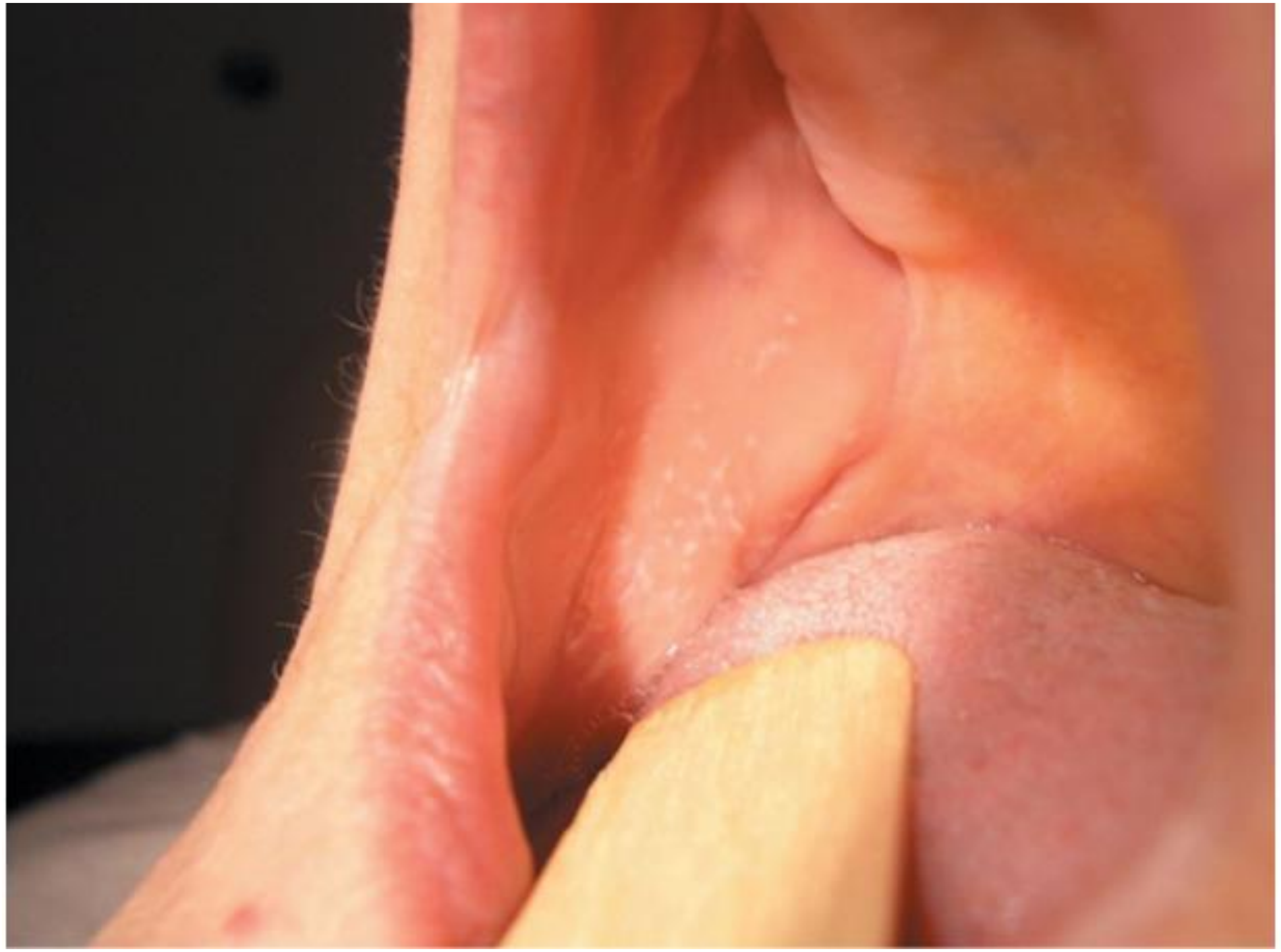
LICHEN PLANUS



paars - zwart
fijn netwerk aan opp. (striae)

50% ook afwijke letsels

hypertrofe
variant




ATOPISCHE DERMATITIS

A

B

A

B

vaak op jonge leeftijd

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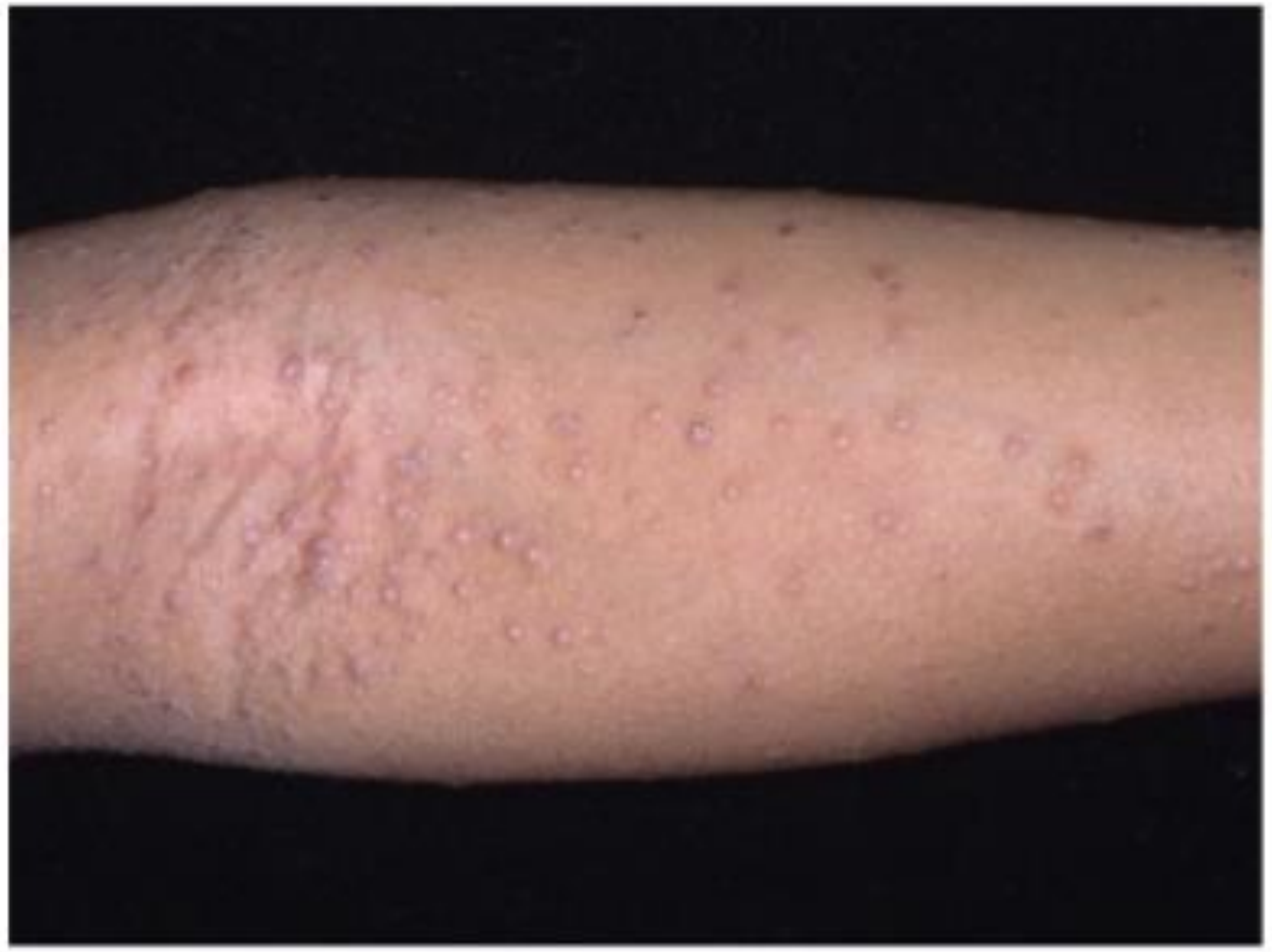
A



B

vaker papillair beeld

DD: mollusca contagiosum

bij genezing vaak hypopigmentatie

max hydratatie

verdikking huid
door eczeem
& krabben

ANDERE VORMEN VAN ECZEEM

dysidrotisch eczeem

allergisch contact eczeem op lippen

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contacteczeem

A

B

chroom - allergie

HERPES ZOSTER

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POLYMORPHOUS LIGHT ERUPTION

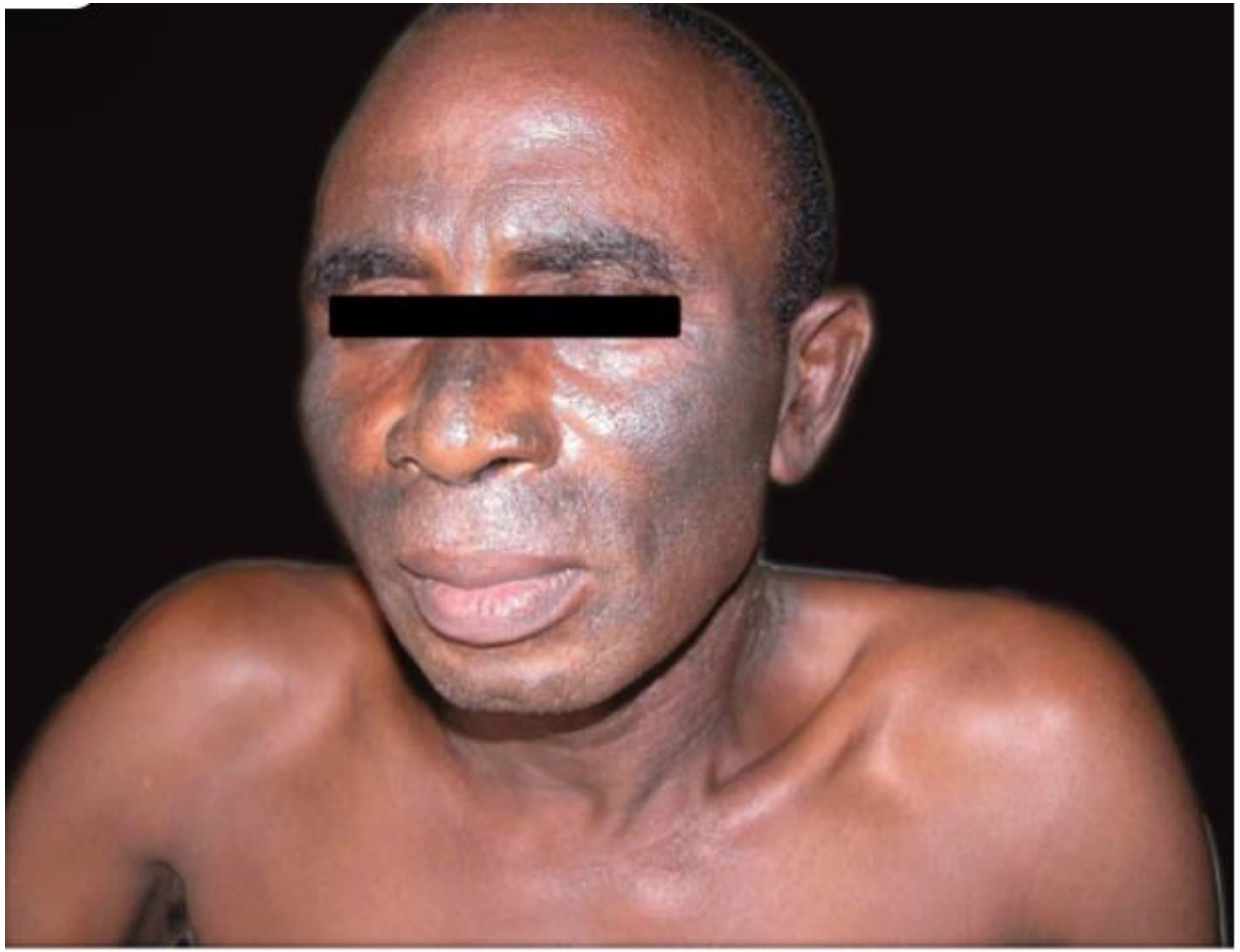
A

B

jeukende huiduitslag door blootstelling (aan UVA)

photodermatose

ACTINISCHE SCHADE/KERATOSEN ~ ook bij type V, VI



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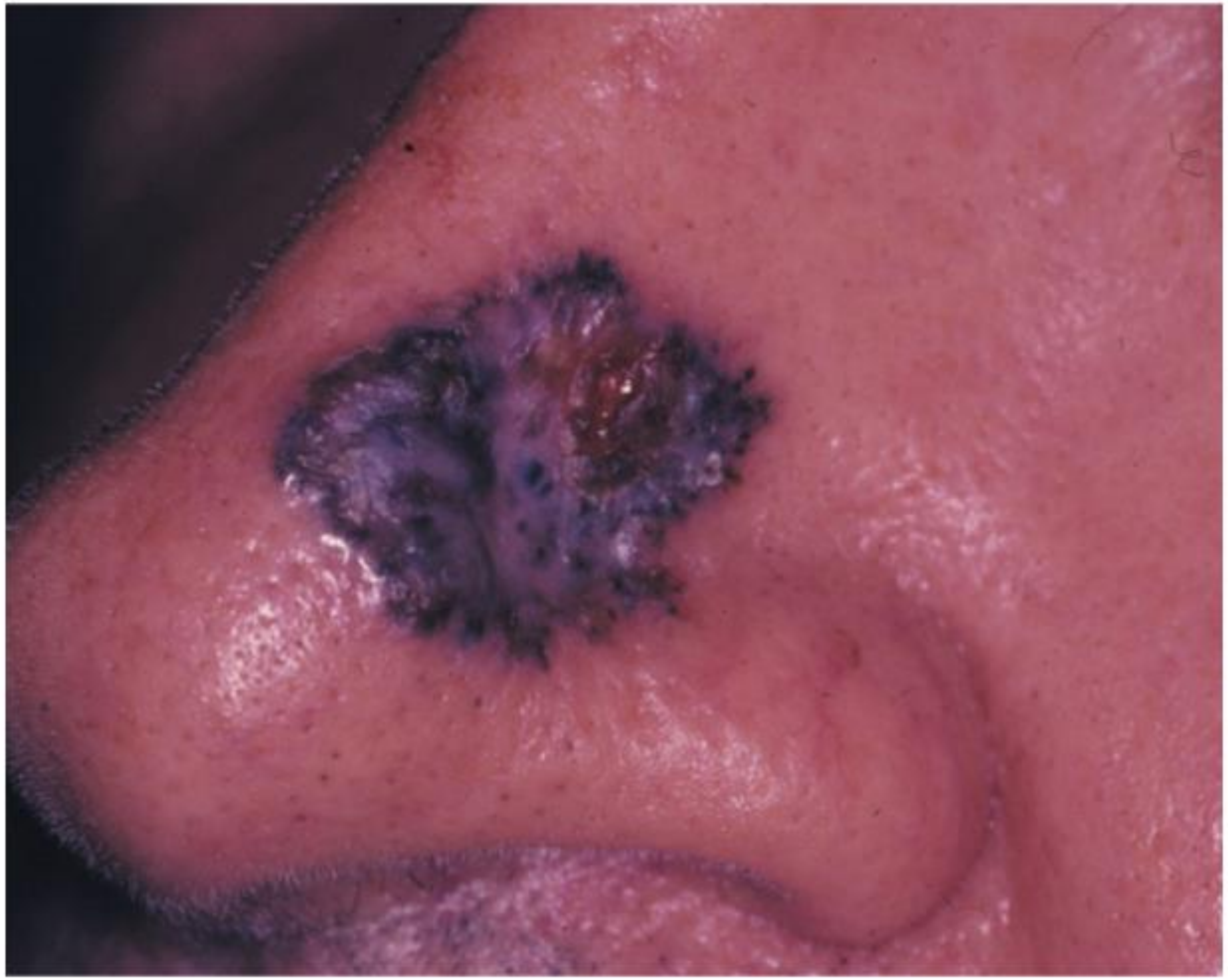
xeroderma pigmentosum ⇝ basocel/spinocellulaire carcinomen

BENIGNE EN MALIGNE HUIDTUMOREN



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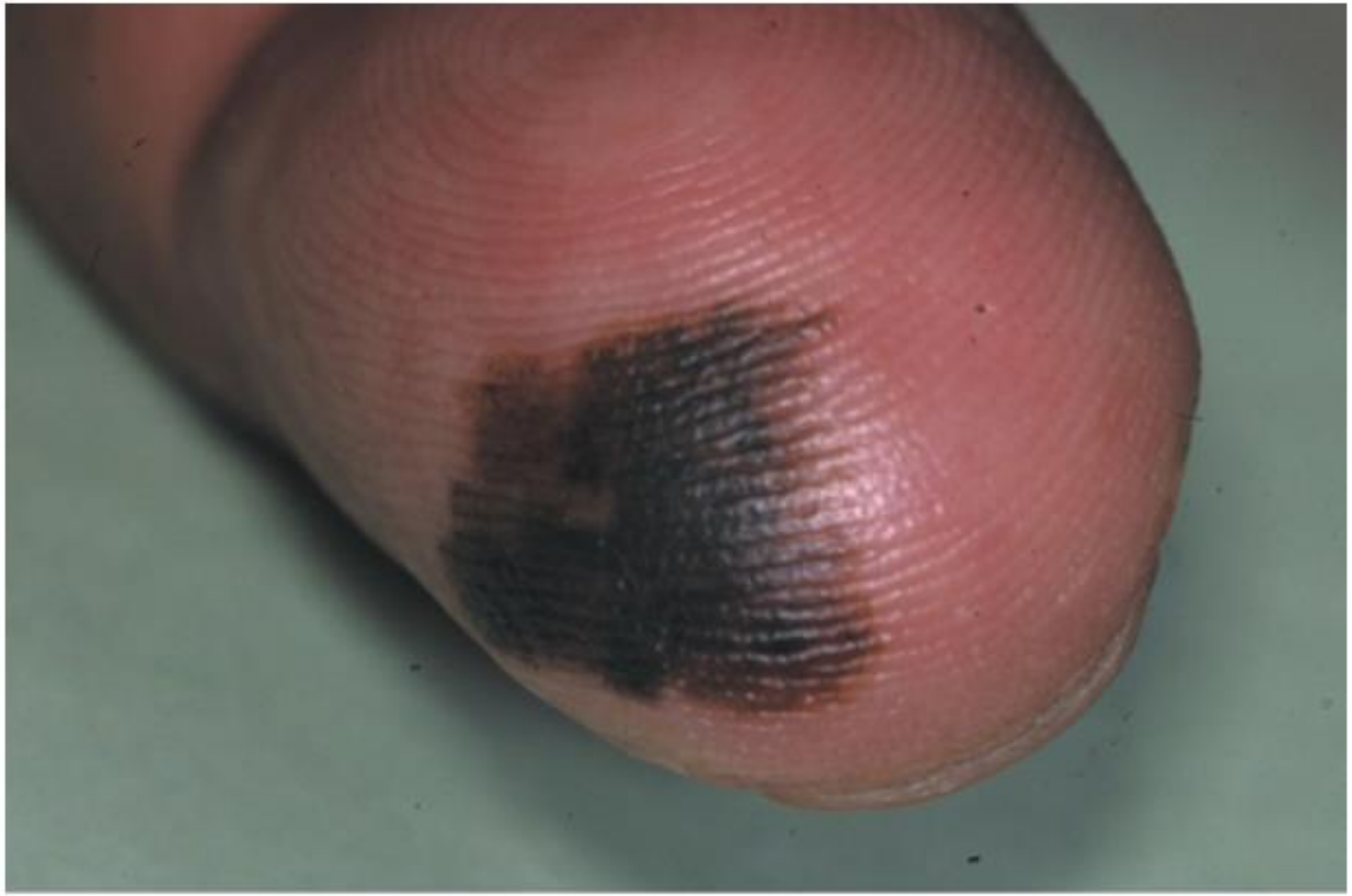

POST INFLAMMATOIRE HYPERPIGMENTATIE

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VITILIGO

MAZELEN

afwezigheid erytheem

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DEEL 4: FYSIOLOGISCHE VERANDERINGEN

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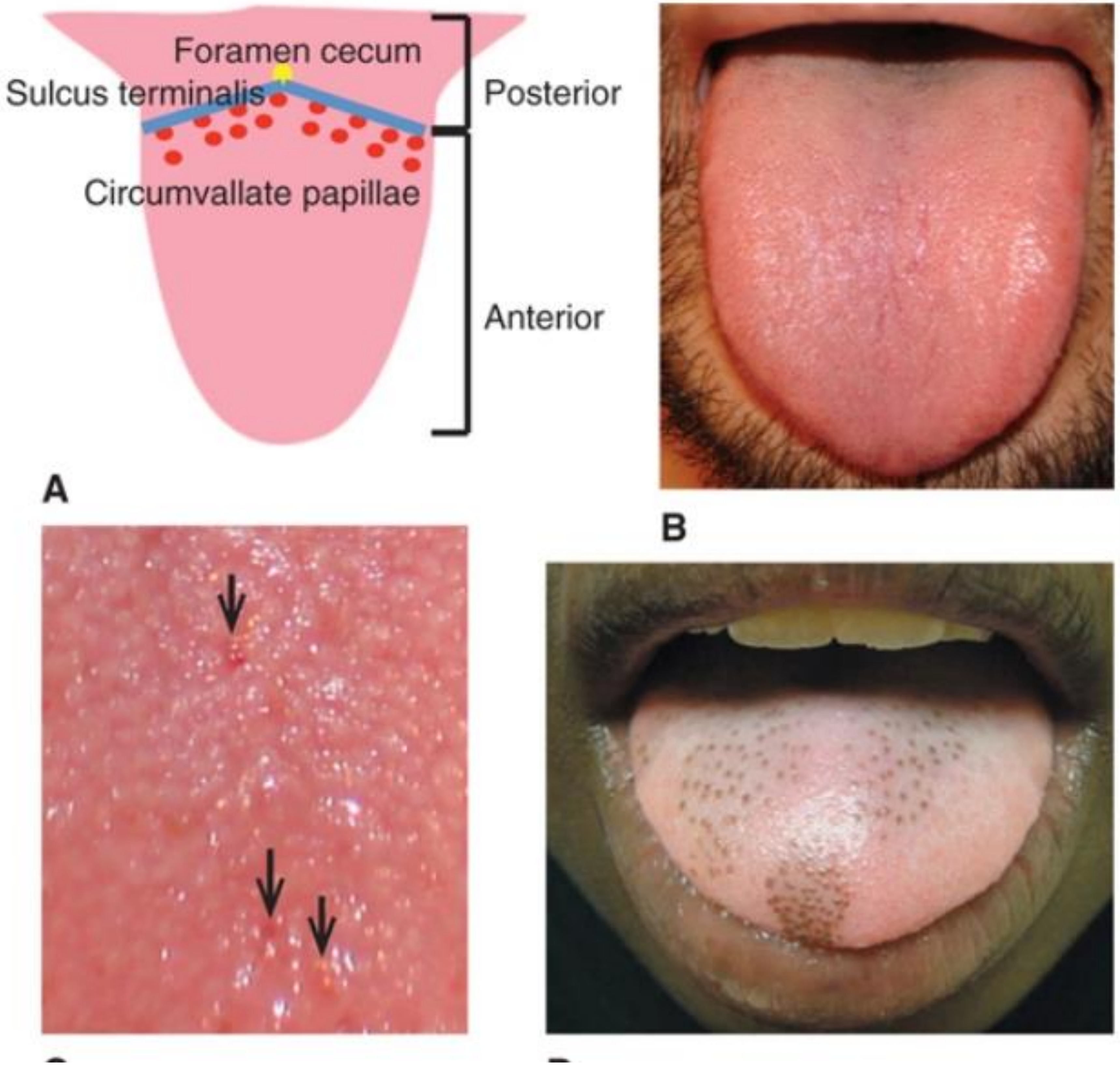

gingivale hyperpigmentatie ook normaal

vaak gepigmenteerde fungiforme papillen

VERSCHILLEN IN PIGMENTATIE

Type	Location	Pigment
A	Anterolateral upper arms, pectoral area	Hyperpigmented
B	Posteromedial aspect of lower legs	Hyperpigmented
C	Vertical line in presternal area	Hypopigmented
D	Posteromedial area of spine	Hyperpigmented
E	Chest from midthird of clavicle to peri-areolar skin	Hypopigmented
F	Straight or curved convex line on the face	Hyperpigmented

A B

Futcher Line

A B

cutane linea alba ook fysiologisch

FACULTEIT GENEESKUNDE EN GEZONDHEIDSWETENSCHAPPEN

Dr. Isabelle Hoorens

Staflid Dermatologie

VAKGROEP HOOFD EN HUID

E isabelle.hoorens@uzgent.be

T +32 9 332 22 17

Universiteit Gent

@ugent

@ugent

Ghent University

www.ugent.be

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